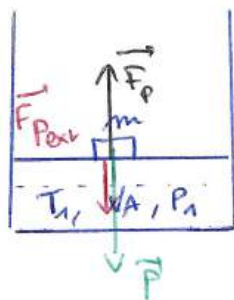
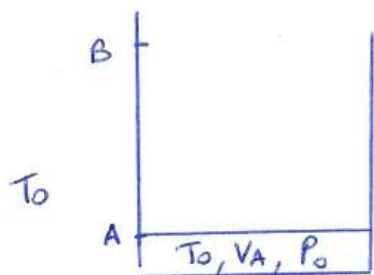


Transformations d'un gaz parfait.



$$C_p - C_v = \frac{(\gamma - 1) nR}{\gamma - 1} = nR$$

$$\textcircled{1} \quad C_p = \frac{\gamma nR}{\gamma - 1}$$

$$C_v = \frac{nR}{\gamma - 1}$$

$\textcircled{2}$ Loi: GP

$$P_0 V_A = nRT_0$$

$$\frac{P_0 V_A}{T_0} = nR$$

$$C_p = \frac{\gamma}{\gamma - 1} \frac{P_0 V_A}{T_0}$$

$$C_v = \frac{1}{\gamma - 1} \frac{P_0 V_A}{T_0}$$

$\textcircled{3}$ Chauffage isochore réversible.

$\textcircled{4}$ Le piston est à l'équilibre $\Rightarrow \sum \vec{F} = \vec{0}$

$$\vec{P} + \vec{F}_{p1} + \vec{F}_{ext} = \vec{0}$$

$$-mg + P_1 S - P_0 S = 0$$

$$P_1 = \frac{P_0 S + mg}{S} = P_0 + \frac{mg}{S}$$

$$\text{Loi GP} \quad P_1 V_A = nRT_1 \Rightarrow T_1 = \frac{P_1 V_A}{nR} = \left(P_0 + \frac{mg}{S} \right) \frac{V_A}{nR}$$

$$T_1 = \left(P_0 + \frac{mg}{S} \right) \cdot \cancel{V_A} \cdot \frac{T_0}{\cancel{P_0 V_A}}$$

$$T_1 = \left(1 + \frac{mg}{SP_0} \right) \cdot T_0 > T_0$$

⑤ 1^{er} principe en gaz de l'enceinte.

$$\Delta U_{0 \rightarrow 1} = W_{0 \rightarrow 1} + Q_{0 \rightarrow 1}$$

$$V = \text{cte} \Rightarrow W_{0 \rightarrow 1} = 0$$

$$Q_{0 \rightarrow 1} = \Delta U_{0 \rightarrow 1}$$

$$Q_{0 \rightarrow 1} = C_V (T_1 - T_0)$$

$$= C_V \Delta T_{0 \rightarrow 1}$$

$$= \frac{nR}{\gamma - 1} (T_1 - T_0)$$

$$\text{or } T_1 - T_0 = \frac{mg}{SP_0} \cdot T_0$$

⑥

$$Q_{0 \rightarrow 1} = \frac{P_0 V_A}{T_0 (\gamma - 1)} \cdot \frac{mg T_0}{SP_0}$$

$$nR = \left(P_0 + \frac{mg}{S} \right) \frac{V_A}{T_1}$$

$$nR = \frac{P_0 V_A}{T_0}$$

$$Q_{01} = \frac{P_0 V_A}{\gamma - 1} \left(\frac{T_1}{T_0} - 1 \right)$$

(7)

dilatation
isobare reversible

$$P_2 = P_1 = P_0 + \frac{mg}{S}$$

(8)

Loi GP

$$P_2 V_B = nRT_2$$

$$P_1 V_A = nRT_1$$

$$P_1 V_B = nRT_2$$

$$\frac{P_1 V_A}{T_1} = nR$$

$$T_2 = \frac{P_1 V_B}{nR} = \frac{P_1 V_B}{P_0 V_A} T_0$$

$$= \frac{\cancel{P_1} \cdot V_B \cdot T_1}{\cancel{P_1} \cdot V_A}$$

$$T_2 = \frac{V_B}{V_A} T_1$$

$$nR(T_2 - T_1) = P_2(V_B - V_A)$$

(9)

Le piston est à l'équilibre, ce qui impose

$$P = P_0 + \frac{mg}{S}$$

ainsi $P_2 = P_1 = P(L)$ isobare.

(10) Isobare: enthalpie

(11)

$$\Delta H_{1 \rightarrow 2} = Q_{1 \rightarrow 2}$$

transformation isobare ($w' = 0$)

$$Q_{12} = C_p(T_2 - T_1) = \frac{nR\gamma}{\gamma - 1}(T_2 - T_1)$$

(12)

$$Q_{12} = \frac{P_0 V_A}{T_0} \cdot \frac{\gamma}{\gamma - 1} (T_2 - T_1)$$

$$nR = \frac{P_0 V_A}{T_0}$$

(13)

$$\Delta u = C_v \Delta T = C_v(T_2 - T_1) = w + Q$$

$$w = \Delta u - Q = C_v \Delta T - C_p \Delta T = -nR \cdot \Delta T$$

$$= -\frac{P_0 V_A}{T_0} (T_2 - T_1)$$

$$= -\cancel{nR} \left(\frac{P_2 V_B}{\cancel{nR}} - \frac{P_1 V_A}{\cancel{nR}} \right)$$

$$w_{12} = -P_1(V_B - V_A)$$

(14) Dans l'état 3, à l'équilibre $P_3 = P_0$.

Refroidissement isochore réversible.

(15) Rechauffement isochore réversible ($P_3 = P_0 = P(t)$)

(16) $W_{23} = 0$ ($dV = 0$)

$$W_{30} = - \int P_{\text{ext}} dV = \int -P dV = -P_0 (V_F - V_I) = -P_0 (V_A - V_B)$$

$$W_{30} = P_0 (V_B - V_A)$$

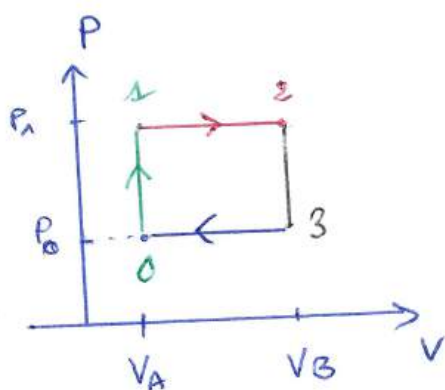
(17) $W_{\text{cycle}} = W_{01} + W_{12} + W_{23} + W_{30}$

$$W_{\text{cycle}} = 0 - P_1 (V_B - V_A) + 0 + P_0 (V_B - V_A)$$

$$= (V_B - V_A) (P_0 - P_1)$$

$$W_{\text{cycle}} = - \frac{mg}{S} \underbrace{(V_B - V_A)}_{>0} < 0 \quad \text{cycle moteur.}$$

(18)



↻ cycle moteur
 $Q > 0, W < 0$

(19)

$$W = -A = (P_1 - P_0)(V_A - V_B)$$